
NOTES FOR TMTS

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1 Useful Conclusions and Fundamental Definitions

1.1 Probability Theory

Theorem 1.1.1 (Kolmogorov's Theorem). The probability distribution functions $(F_{\mathbf{t}})_{\mathbf{t} \in \mathcal{T}}$ are the distribution functions of some stochastic process if and only if for any integer n and $\mathbf{t} = (t_1, \dots, t_n) \in \mathcal{T}$ and $1 \leq i \leq n$,

$$\lim_{x_i \rightarrow \infty} F_{\mathbf{t}}(x) = F_{\mathbf{t}(i)}(x(i))$$

where $\mathbf{t}(i)$ and $x(i)$ are the $(n-1)$ -component vectors obtained by deleting the i th components of $\mathbf{t}$ and x respectively.

Theorem 1.1.2 (Spectral Theorem for Compact Self-Adjoint Operators). Let H be a real or complex Hilbert space, and let $T : H \rightarrow H$ be a compact self-adjoint operator. Then there exists an orthonormal family $\{e_n\}$ consisting of eigenvectors of T corresponding to nonzero eigenvalues $\{\lambda_n\}$ such that

$$H = \ker T \oplus \overline{\text{span}\{e_n\}}.$$

Moreover, the following properties hold:

1. Every eigenvalue λ_n is real.
2. Every nonzero eigenspace is finite-dimensional.
3. The set of nonzero eigenvalues is at most countable, and if there are infinitely many of them, then 0 is their only possible accumulation point.
4. For every $x \in H$,

$$Tx = \sum_n \lambda_n \langle x, e_n \rangle e_n.$$

In particular, if $\ker T = \{0\}$, then $\{e_n\}$ is an orthonormal basis of H .

Proof. See Appendix (A.1.1). □

1.2 Definitions

Definiton 1.2.1 (Stationarity). The time series $(X_t)_{t \in \mathbb{Z}}$ is said to be stationary if

- $\mathbb{E}|X_t|^2 < \infty$ for all $t \in \mathbb{Z}$.
- $\mathbb{E}X_t = m$ for all $t \in \mathbb{Z}$.
- $\text{Cov}(X_s, X_t) =: \gamma_X(s, t) = \gamma_X(s + h, t + h)$ for all $s, t, h \in \mathbb{Z}$.

Definiton 1.2.2 (Strict Stationarity). The time series $(X_t)_{t \in \mathbb{Z}}$ is said to be strictly stationary if the joint distributions of $(X_{t_1}, \dots, X_{t_k})$ and $(X_{t_1+h}, \dots, X_{t_k+h})$ are the same for all positive integers k and for all $t_1, \dots, t_k, h \in \mathbb{Z}$.

The strict stationarity may imply the stationarity easily.

Definiton 1.2.3 (Elementary Properties). . For a stationary process, we define the auto-covariance $\gamma_X(h) = \gamma_X(t+h, t)$ for any $t \in \mathbb{Z}$ which is well-defined.

If γ is the autocovariance function of a stationary process $(X_t)_{t \in \mathbb{Z}}$, then

$$\gamma(0) \geq 0, \quad |\gamma(h)| \leq \gamma(0) \text{ for all } h \in \mathbb{Z}$$

and $\gamma(h) = \gamma(-h)$ for any $h \in \mathbb{Z}$.

Proof. We only prove that $\gamma(0) \geq |\gamma(h)|$ for any $h \in \mathbb{Z}$.

$$\begin{aligned} \gamma(h)^2 &= (\mathbb{E}(X_0 - \mu(0))(X_h - \mu(h)))^2 \\ &\leq \mathbb{E}(X_0 - \mu(0))^2 \mathbb{E}(X_h - \mu(h))^2 = \gamma(0)^2 \end{aligned}$$

by Cauchy-Schwarz Inequality. □

Definiton 1.2.4 (Non-Negative Definiteness). A real-valued function on the integers, $\kappa : \mathbb{Z} \rightarrow \mathbb{R}$ is said to be non-negative definite if and only if

$$\sum_{i,j=1}^n a_i \kappa(t_i, t_j) a_j \geq 0$$

for all positive integers n and for all vectors $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ and $t = (t_1, \dots, t_n) \in \mathbb{Z}$.

Remark. Replace t_i with i in the definition above will obtain an equivalent conclusion.

Theorem 1.2.1 (Characterization of Autocovariance Functions). A real-valued function defined on the integers is the autocovariance function of a stationary time series if and only if it is even and non-negative definite.

Definiton 1.2.5 (Sample Autocovariance Function). The sample autocovariance function of $\{x_1, \dots, x_n\}$ is defined by

$$\hat{\gamma}(h) := \frac{1}{n} \sum_{i=1}^{n-h} (x_i - \bar{x})(x_{i+h} - \bar{x}), 0 \leq h < n$$

and $\gamma(h) = \gamma(-h)$, $-n < h \leq 0$.

Proposition 1.2.2. The covariance matrix is symmetric and non-negative definite for any random vector.

Proposition 1.2.3. If $\mathbf{a}$ is an m -component column vector, B is an $m \times n$ matrix, and $\mathbf{X} = (X_1, \dots, X_n)$ where $\mathbb{E}|X_i|^2 < \infty$, $i = 1, \dots, n$, then the random vector,

$$\mathbf{Y} = \mathbf{a} + B\mathbf{X}$$

has the mean $\mathbb{E}\mathbf{Y} = \mathbf{a} + B\mathbb{E}\mathbf{X}$ and covariance matrix

$$\Sigma_{\mathbf{Y}\mathbf{Y}} = B\Sigma_{\mathbf{X}\mathbf{X}}B^T$$

Definiton 1.2.6 (Multivariate Normal Distribution). The random vector $\mathbf{Y} = (Y_1, \dots, Y_n)$ is said to be multivariate normal, or to have a multivariate normal distribution, if and only if there exist a column vector $\mathbf{a}$, a matrix B and a random vector $\mathbf{X} = (X_1, \dots, X_m)$ with

independent standard normal components, such that $\mathbf{Y} = \mathbf{a} + B\mathbf{X}$, with

$$\mathbb{E}\mathbf{Y} = \mathbf{a}, \quad \Sigma\mathbf{Y} = BB^T$$

Proposition 1.2.4. If $\mathbf{Y} = (Y_1, \dots, Y_n)$ is a multivariate normal random vector such that $\mathbb{E}\mathbf{Y} = \mu$ and $\Sigma_{\mathbf{Y}\mathbf{Y}} = \Sigma$, then the characteristic function of $\mathbf{Y}$ is

$$\phi_{\mathbf{Y}}(\mathbf{x}) = \exp\left(i\mathbf{x}^T\mu - \frac{1}{2}\mathbf{x}^T\Sigma\mathbf{x}\right)$$

If $\det \Sigma > 0$ then $\mathbf{Y}$ has the density

$$f_{\mathbf{Y}}(\mathbf{y}) = (2\pi)^{-n/2}(\det \Sigma)^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{y} - \mu)' \Sigma^{-1}(\mathbf{y} - \mu)\right]. \quad (1.6.12)$$

Proof. Consider $\mathbf{X}$ to be an independent standard normal vector, then we know $\mu + \Sigma^{1/2}\mathbf{X} \stackrel{d}{=} \mathbf{Y}$, then

$$\begin{aligned} \phi_{\mathbf{Y}}(\mathbf{x}) &= \mathbb{E}e^{i\mathbf{x}^T\mathbf{Y}} = \mathbb{E}e^{i\mathbf{x}^T(\mu + \Sigma^{1/2}\mathbf{X})} \\ &= \exp(i\mathbf{x}^T\mu)\phi_{\mathbf{X}}(\Sigma^{1/2}\mathbf{x}) = \exp\left(i\mathbf{x}^T\mu - \frac{1}{2}\mathbf{x}^T\Sigma\mathbf{x}\right) \end{aligned}$$

□

2 Stationary ARMA Processes

2.1 ARMA Processes

Definiton 2.1.1 (White Noise). The process (Z_t) is said to be white noise with mean 0 and variance σ^2 if and only if (Z_t) has zero mean and covariance function

$$\gamma(h) = \begin{cases} 0, & h \neq 0 \\ \sigma^2, & h = 0 \end{cases}$$

Definiton 2.1.2 (The ARMA(p, q) Process). The process $(X_t)_{t \in \mathbb{Z}}$ is said to be an ARMA- (p, q) process if (X_t) is stationary and if for every t ,

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$$

where $(Z_t) \sim \text{WN}(0, \sigma^2)$. We say that (X_t) is an ARMA(p, q) process with mean μ if $(X_t - \mu)$ is an ARMA(p, q) process.

The equation can be written as

$$\phi(B)X_t = \theta(B)Z_t$$

where ϕ, θ are p, q -degree polynomials and B is the push-backward operator.

Definiton 2.1.3 (The MA(q) process and the AR(p) process). The MA(q) process is an ARMA($0, q$) process, that is $\phi \equiv 1$, i.e.

$$X_t = \theta(B)Z_t.$$

The AR(p) process is an ARMA($p, 0$) process, that is $\theta \equiv 1$, i.e.

$$\phi(B)X_t = Z_t$$

Proposition 2.1.1. The mean value and covariance function for an MA(q) process X_t with MA polynomial θ is

$$\mu_X(t) = 0, \quad \gamma_X(h) = \sigma^2 \left(\sum_{k=0}^{q-h} \theta_k \theta_{k+h} \right).$$

Proof. We know

$$\mathbb{E}X_t = \mathbb{E}\theta(B)Z_t = 0$$

and

$$\gamma_X(h) = \mathbb{E}(\mathbb{E}\theta(Z)Z_t)(\theta(Z)Z_{t+h}) = \sigma^2 \left(\sum_{k=0}^{q-h} \theta_k \theta_{k+h} \right)$$

where $\theta_0 := 1$. □

Definiton 2.1.4 (Causality). An ARMA(p, q) process defined by the equations $\phi(B)X_t = \theta(B)Z_t$ is said to be *causal* (or more specifically to be a *causal function of* $\{Z_t\}$) if there

exists a sequence of constants $\{\psi_j\}_{j \geq 0}$ such that

$$\sum_{j=0}^{\infty} |\psi_j| < \infty$$

and

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \quad t = 0, \pm 1, \dots$$

Proposition 2.1.2. If $\{X_t\}_{t \in \mathbb{Z}}$ is any sequence of random variables such that $\sup_t \mathbb{E}|X_t| < \infty$, and if $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, then the series

$$\psi(B)X_t = \sum_{j=-\infty}^{\infty} \psi_j B^j X_t = \sum_{j=-\infty}^{\infty} \psi_j X_{t-j}$$

converges absolutely with probability one. If in addition $\sup_t \mathbb{E}|X_t|^2 < \infty$, then the series converges in mean square to the same limit.

Proof. To see the absolute convergence for a.s., we may notice that

$$\mathbb{E}|\phi(B)|X_t| \leq \sum_{j=-\infty}^{\infty} |\phi_j| \sup \mathbb{E}|X_t| < \infty$$

and hence $|\phi(B)|X_t|$ has to be finite in probability 1.

Then we would like to show that the series converge to $\phi(B)X_t$ in L^2 as well, that is we may prove that by noticing

$$\mathbb{E} \left| \sum_{m < |j| \leq n} \phi_j X_{t-j} \right|^2 \leq \left(\sum_{|j| \geq m} |\phi_j| \right)^2 \sup_t \mathbb{E}|X_t|^2 \rightarrow 0$$

and hence the series converge in L^2 to some limit L , with

$$\mathbb{E}|L - \sum_{j=-\infty}^{\infty} \phi_j X_{t-j}|^2 \leq \liminf \mathbb{E}|L - \sum_{j=-n}^n \phi_j X_{t-j}|^2 = 0$$

by Fatou's lemma and we are done. $\square$

Proposition 2.1.3. If $\{X_t\}$ is a stationary process with autocovariance function $\gamma(\cdot)$ and if $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, then for each $t \in \mathbb{Z}$ the series

$$\psi(B)X_t = \sum_{j=-\infty}^{\infty} \psi_j X_{t-j}$$

converges absolutely with probability one and in mean square to the same limit. If

$$Y_t = \psi(B)X_t,$$

then the process $\{Y_t\}$ is stationary with autocovariance function

$$\gamma_Y(h) = \sum_{j,k=-\infty}^{\infty} \psi_j \psi_k \gamma(h-j+k).$$

Theorem 2.1.4. Let $\{X_t\}$ be an ARMA(p, q) process for which the polynomials $\phi(\cdot)$ and $\theta(\cdot)$ have no common zeroes. Then $\{X_t\}$ is causal if and only if $\phi(z) \neq 0$ for all $z \in \mathbb{C}$ such that $|z| \leq 1$. The coefficients $\{\psi_j\}$ in the causal representation $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ are determined by the relation

$$\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\theta(z)}{\phi(z)}, \quad |z| \leq 1.$$

Proof. Suppose first that $\phi(z) \neq 0$ for every $|z| \leq 1$. Since the zero set of the polynomial ϕ is finite, there is some $R > 1$ such that ϕ has no zero in $|z| < R$. Hence the rational function

$$\psi(z) := \frac{\theta(z)}{\phi(z)}$$

is analytic in $|z| < R$ and has a power-series expansion

$$\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j.$$

Choose r with $1 < r < R$. Cauchy's estimate gives

$$|\psi_j| \leq M_r r^{-j}, \quad M_r := \max_{|z|=r} |\psi(z)|,$$

so $\sum_{j=0}^{\infty} |\psi_j| < \infty$. Consequently

$$Y_t := \psi(B)Z_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

is well defined, and the preceding propositions show that $\{Y_t\}$ is stationary. Since $\phi(z)\psi(z) = \theta(z)$, absolute summability permits composition of the filters and gives

$$\phi(B)Y_t = \theta(B)Z_t = \phi(B)X_t.$$

Put $U_t = X_t - Y_t$. Then $\{U_t\}$ is stationary and $\phi(B)U_t = 0$. The function $1/\phi(z)$ is also analytic in $|z| < R$ and has absolutely summable Taylor coefficients. Applying the corresponding filter to the last equality yields

$$U_t = \frac{1}{\phi}(B)\phi(B)U_t = 0.$$

Thus $X_t = Y_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$, so $\{X_t\}$ is causal.

Conversely, suppose that $\{X_t\}$ is causal, so that

$$X_t = \psi(B)Z_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \quad \sum_{j=0}^{\infty} |\psi_j| < \infty.$$

Substitution into the ARMA equation gives

$$(\phi(B)\psi(B) - \theta(B))Z_t = 0.$$

Write

$$\phi(z)\psi(z) - \theta(z) = \sum_{j=0}^{\infty} a_j z^j.$$

The sequence $\{a_j\}$ is absolutely summable. Since $\{Z_t\}$ is white noise with variance $\sigma^2 > 0$,

$$0 = \text{Var} \left(\sum_{j=0}^{\infty} a_j Z_{t-j} \right) = \sigma^2 \sum_{j=0}^{\infty} |a_j|^2.$$

Therefore $a_j = 0$ for every j , and hence

$$\phi(z)\psi(z) = \theta(z), \quad |z| \leq 1.$$

If $\phi(z_0) = 0$ for some $|z_0| \leq 1$, the absolute convergence of $\psi(z_0)$ would imply $\theta(z_0) = 0$, contradicting the assumption that ϕ and θ have no common zeroes. Thus $\phi(z) \neq 0$ throughout $|z| \leq 1$, and the displayed identity also gives $\psi(z) = \theta(z)/\phi(z)$. This relation uniquely determines all of the coefficients $\{\psi_j\}$. $\square$

Definiton 2.1.5 (Invertibility). An ARMA(p, q) process defined by the equations $\phi(B)X_t = \theta(B)Z_t$ is said to be *invertible* if there exists a sequence of constants $\{\pi_j\}_{j \geq 0}$ such that

$$\sum_{j=0}^{\infty} |\pi_j| < \infty$$

and

$$Z_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}, \quad t = 0, \pm 1, \dots$$

Theorem 2.1.5. Let $\{X_t\}$ be an ARMA(p, q) process for which the polynomials $\phi(\cdot)$ and $\theta(\cdot)$ have no common zeroes. Then $\{X_t\}$ is invertible if and only if $\theta(z) \neq 0$ for all $z \in \mathbb{C}$ such that $|z| \leq 1$. The coefficients $\{\pi_j\}$ in the invertible representation $Z_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}$ are determined by the relation

$$\pi(z) = \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad |z| \leq 1.$$

The coefficients $\{\pi_j\}$ can be calculated from recursion relations analogous to those for $\{\psi_j\}$.

A

A.1

Theorem A.1.1 (Proof for the Spectral Thoerem).

Proof. First notice that

$$\|T\| = \sup_{\|x\| \leq 1} |\langle Tx, x \rangle|,$$

for the proof, let us denote the supremum on the right side to be M , then we know for $x, y \in H$, we have

$$\langle T(x+y), x+y \rangle = \langle Tx, x \rangle + \langle Ty, y \rangle + 2\operatorname{Re}(\langle Tx, y \rangle)$$

and hence

$$\langle T(x+y), x+y \rangle - \langle T(x-y), x-y \rangle = 4\operatorname{Re}(\langle Tx, y \rangle)$$

which gives

$$|4\operatorname{Re}(\langle Tx, y \rangle)| \leq M(\|x+y\|^2 + \|x-y\|^2).$$

We will have

$$4\|Tx\| \leq 4M$$

with $y = (Tx)^0$ and $\|x\| \leq 1$. Therefore, the equality goes and with this, there exists x_n in the unit ball such that

$$\langle Tx_n, x_n \rangle \rightarrow M$$

then we know there is a cumulation x of $\{\pm x_n\}$ such that

$$\langle Tx, x \rangle = M \implies \|Tx - Mx\|^2 \leq 2M^2 - 2M^2 = 0$$

and hence $Tx = Mx$ which means there is a non-zero eigenvector of T . It is easy to check that every eigenvalue is real and eigenvector with distinct eigenvalue is orthogonal. To see that an eigenspace is finite-dimensional, assume that $Te_i = \lambda e_i$ for $i \in I$ for a infinite index set I . We may construct an orthonormal sequence of e_i and we know $\|Te_i - Te_j\| = \sqrt{2}|\lambda|$ which is a contradiction to T compact, so an eigenspace has to be finite-dimensional. Similarly, for any $\delta > 0$, the eigenvalues of T larger than δ has to be finite and hence they are at most countable with only 0 possible accumulation point.

Now we consider $M := \overline{\operatorname{span}\{e, Te = \lambda e, \lambda \neq 0\}}$ which is a close subspace of H with $H = M^\perp \oplus M$. Then for any $x \in M^\perp$, we know $Tx \in M^\perp$ and hence $T|_{M^\perp}$ is still adjoint compact and hence it has an eigenvector with non-zero eigenvalue if $\|T|_{M^\perp}\| > 0$, which is a contradiction. Therefore, $M^\perp \subset \ker T$, and easy to check the opposite direction and we are done. $\square$